PGP-DSBA Online- May A Batch

Time Series Analysis

Business Report

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1 Problem 1: Time Series Analysis

For this particular assignment, the data of different types of wine sales in the 20th century is to be analysed. Both of these data are from the same company but of different wines. As an analyst in the ABC Estate Wines, you are tasked to analyse and forecast Wine Sales in the 20th century.

1.1 Read the data as an appropriate Time Series data and plot the data.

The given data are the sales of Rose and Sparkling wine between the year January 1980 to July 1995. With the monthly interval meaning one entry representing the total sales on that month. Hence, we have in total 187 observations or entries for the data.

The column YearMonth is not in the date format and hence, the same is dropped and a date range equivalent to the timeframe of the data is incorporated and the same is converted as index.

Figure 2: Spark Data_Yearly

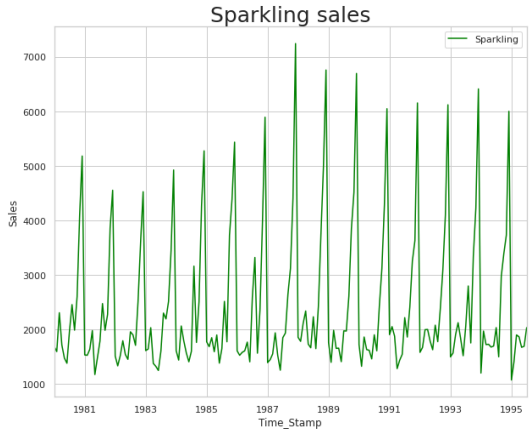
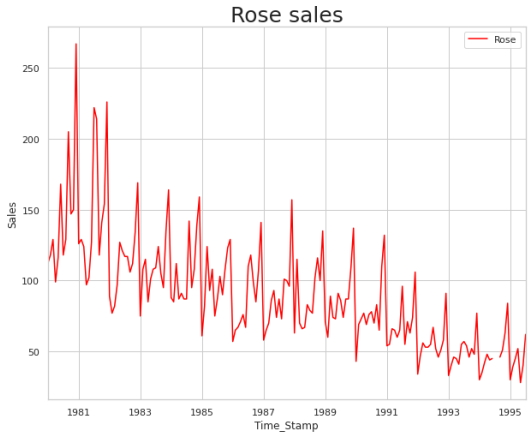


Figure 1 Rose Data_Yearly



While looking at the Figure 2 and Figure 1 in the data, we could see there is a hybrid trend in the Sales of the Sparkling and on the other hand There is a stable drop in sales in Rose. Both the data exhibit some kind of seasonal pattern which has some local peaks each year. Let us explore it in the upcoming sessions.

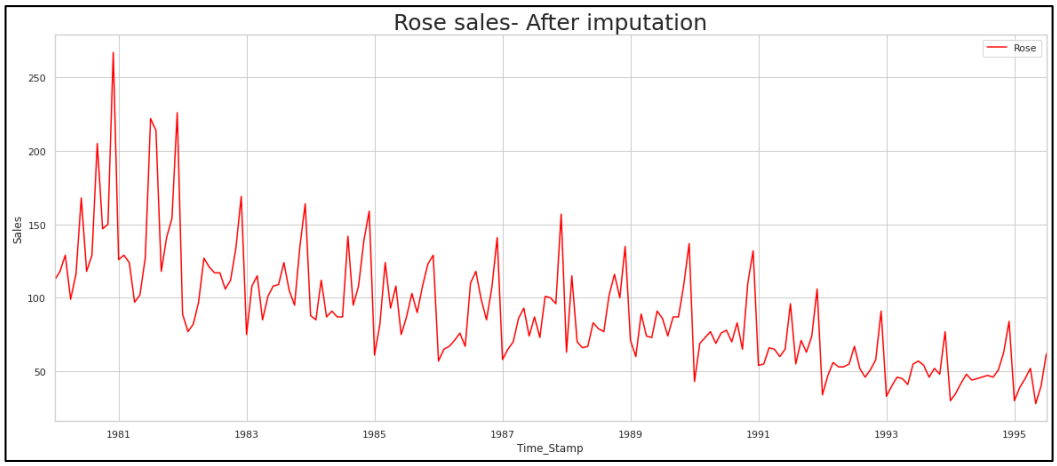
Before going further into the Time series Analysis, it is imperative that we should have a contiguous data set. Since the time series data is order sensitive and the data is independent, blank/ null values in the dataset are not acceptable. As it may lead to change in the data structure completely. Hence suitable imputation has to be done for the null values so that we can bring the data as a contiguous data.

Figure 3: Missing Data in Rose

	Sparkling	Rose
Time_Stamp		
1994-07-31	2031	NaN
1994-08-31	1495	NaN

In our case, If we refer to Figure 3, we could see that Rose has a missing values in 1994. There are two pieces of data missing. And by using the interpolate method- Spline to synthetically insert the data into the data frame. However this is not a random imputation. The value is imputed based on the neighbouring values of the missing data.

Figure 4: Data After null value imputation



With the imputed value the data looks as shown in Figure 4.

1.2 Perform appropriate Exploratory Data Analysis to understand the data and perform decomposition

1.2.1 EDA

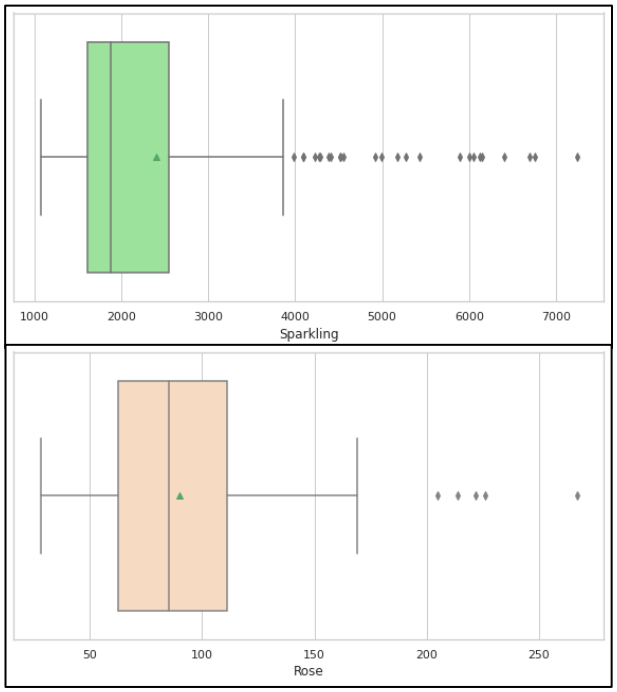
1.2.1.1 Describe Function

Beginning with the describe function, for both the data side by side in Figure 5. We could see that

- Rose data is significantly smaller in scale compared with the sparking data.
- Sparkling data has several outliers in the Although the Wiskers are only until around 3900 the outliers are stretched until 7200
- However, the Coefficient of Variation (CV) appears to be smaller for Sparkling at 1.85 compared to Rose at 2.29. This may be due to the fact the scales of both the data are different.
- The sparkling data appears right skewed and the Rose data is not so much skewed.
- Mean of each data (▲) is also plotted in the Boxplot, it is evident that they are all away from the median. Hence both the data are not normally distrusted.

Figure 5: Descriptive Details Combined

	Sparkling	Rose
count	187.000000	187.000000
mean	2402.417112	89.927152
std	1295.111540	39.224081
min	1070.000000	28.000000
25%	1605.000000	62.500000
50%	1874.000000	85.000000
75%	2549.000000	111.000000
max	7242.000000	267.000000
CV	1.854989	2.292652

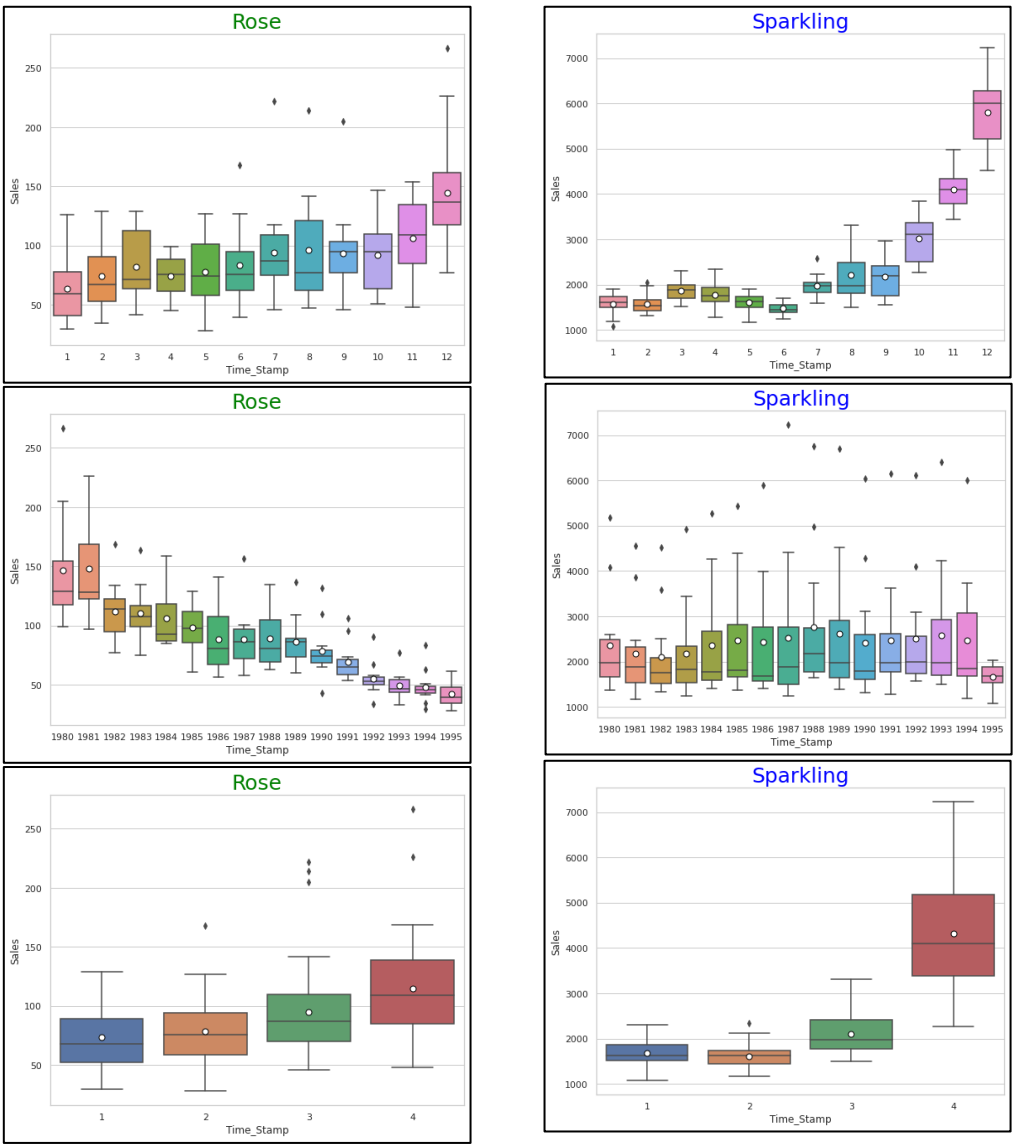


1.2.1.2 Boxplot analysis.

Continuing further with the boxplot of data in Figure 6 to understand their distribution on month year and quarter basis below are the observations.

- Rose Data
  - From the monthly boxplot it is evident that last two months of the year has a significant increase in the overall wine sales. Further the lowest mean recorded is on January.
  - From the yearly boxplot, it is evident that the total sales is diminishing year on year. Record lowest mean sales in 1994. Considering only half the data for 1995 was available this can be ignored. Most of the outliers are concentrated in the recent past only (from 1989).
  - Similar to the monthly boxplot. The quarterly data exhibit higher sales in Q4 as November and December falls on 4th Quarter. Mean represented in white circle is always higher than the median on the quarterly graph.

Figure 6: Box Plot of the Spark and Rose Data (3x2)



- Sparkling Data.
    - From the monthly boxplot data, it is evident that there is significant increment sales after 9th Month, marking the festivity season expected in these three months. The means in these months are always, smaller compared to median. The lowest sales recorded is on June.
    - From the yearly boxplot, it is evident that mean of all the years are greater than the median except 1995 this might be due to the fact that we have partial data for the year 1995. Unlike Rose Data the outliers are seen in all the years. Highest mean recorded in 1988
    - Similar to the monthly boxplot. The quarterly data exhibit significantly very high sales in Q4. The first and second quarter shows similar trend and mean is closer to median.

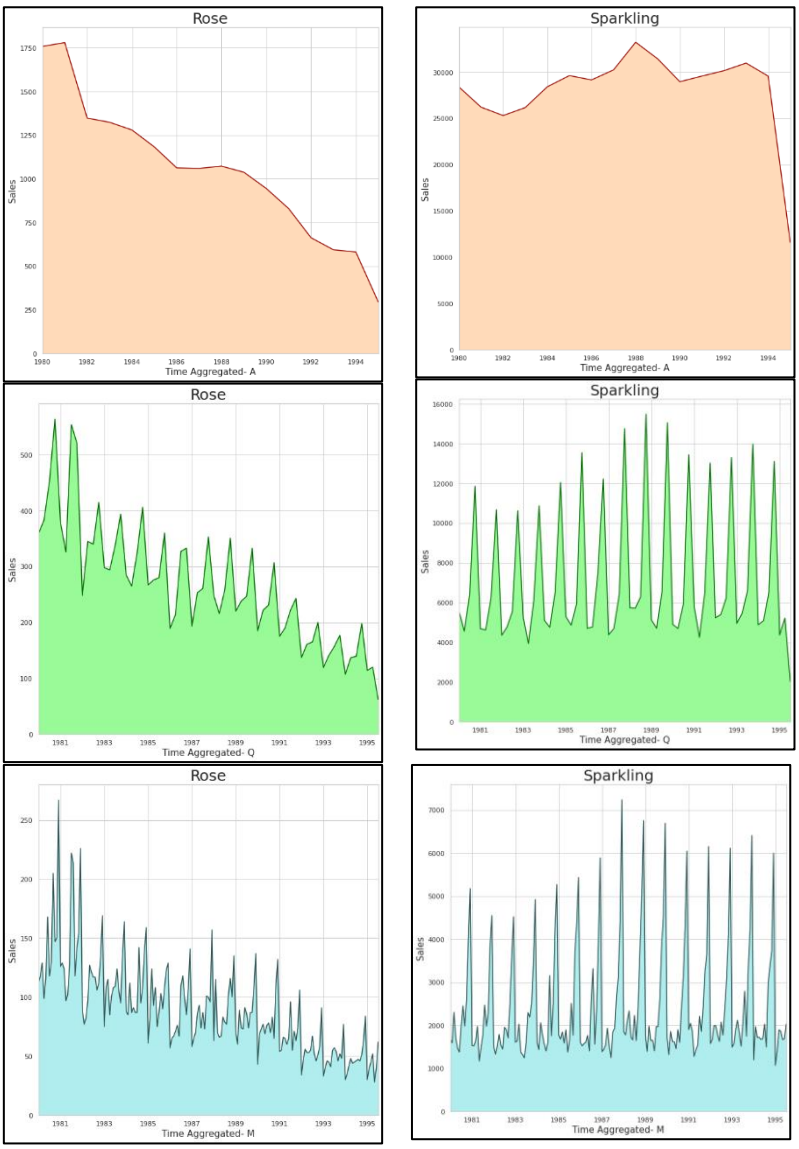
1.2.1.3 Resampling Analysis

Further to the boxplot, resampling would reveal certain trend in the data as shown in Figure 7.

- Rose Data
    - It is seen that from the yearly resampling there is a spike right in the second year and after that there is consistent drop in sales.

- For the quarterly resampling we could see a single spike accompanying with a smaller growth just before the spike shows there are two quarters (Q3 and Q4 has good sales). Evident from the boxplot as well.
- Further downsampling for the monthly data, we could see the same trend following but on the monthly level. In all the levels the over all sales are diminishing as the year progresses.

Figure 7: Resampling of the Spark and Rose data (3x2)



- Spark Data
  - The sales of the spark data on the yearly resampling exhibits no such upward or downward tend but a hybrid trend. There is a Dip in 1982 and 1990.
  - Unlike the Rose data, the spark has a single peak each year in the quarterly data as Q4 sales is very much significant compared to other two.
  - The monthly data as well exhibit the higher trend only in Demeter contributing highest.

1.2.2 Decomposition.

Decomposition of the data reveals the components of the series such as trend, seasonality and the residual.

- The trend is the overall movement of the data in the given span of time.
- Seasonality is the repetitive pattern observed in the data within a year.
- Residuals are the unexplainable part of the data which are caused due to other external factor which is not repeatable.

We can decompose the data in two ways multiplicative and additive.

- When the seasonal component of the data has an increasing magnitude as the trend increases, then we can do multiplicative decomposition.
- When the seasonal component is relatively similar thorough out the time series then we can take additive decomposition.

Looking the data as such, sometime provides the information regarding how the decomposition can be done. If not we can perform Additive decomposition. If there are any pattern observed in the residual then we can do the multiplicative Decomposition.

1.2.2.1 Rose Data

While performing the Additive decomposition for the Rose data in Figure 8, the residual appears to be having peak and trough. This is not in a repetitive pattern. Even if tried with multiplicative decomposition there is no significant change in the residual.

Figure 8: Decomposed Data- Rose

1.2.2.2 Spark

While performing the Additive decomposition for the Spark data in Figure 9, the residual appears to be having peak and trough. This is not in a repetitive pattern. Even if tried with multiplicative decomposition there is no significant change in the residual.

Figure 9: Decomposed Data- Spark

1.3 Split the data into training and test. The test data should start in 1991.

Since the order of the time series data to be respected at any given case, care should be taken while performing splitting. The test train data should be split in the orderly manner. In the given case, the data should be split in such a way the train data should have all the entries until end of 1990 and test data should have entries from 1991.

It is also imperative, if there is a seasonal component in the data, the test data should have at least 1 seasonal component in it so as to predict the seasonality. However, in our case the seasonality is well within when we split between 1991, we have covered the seasonal data as well.

The head information of both the data is shown Figure 11, Figure 10.

Figure 11: Train Test- Rose

```
Train size is (132, 1)
Test size is (55, 1)
             Rose
Time_Stamp
1980-01-31  112.0
1980-02-29  118.0
1980-03-31  129.0
1980-04-30   99.0
1980-05-31  116.0
            Rose
Time_Stamp
1991-01-31  54.0
1991-02-28  55.0
1991-03-31  66.0
1991-04-30  65.0
1991-05-31  60.0
```

Figure 10 Train Test- Sparkling

```
Train size is (132, 1)
Test size is (55, 1)
            Sparkling
Time_Stamp
1980-01-31       1686
1980-02-29       1591
1980-03-31       2304
1980-04-30       1712
1980-05-31       1471
            Sparkling
Time_Stamp
1991-01-31       1902
1991-02-28       2049
1991-03-31       1874
1991-04-30       1279
1991-05-31       1432
```

1.4 Build various exponential smoothing models on the training data and evaluate the model using RMSE on the test data. Other models such as regression,naïve forecast models and simple average models. should also be built on the training data and check the performance on the test data using RMSE..

1.4.1 Exponential Smoothing.

Exponential smoothing is the method where the weightage is given for the past data based on its recency.

If the data is recent then the weightage is given higher considering it will have a higher impact on the data, however for the older data we will have a smaller weightage. This delay in weightage is controlled by a parameter.

- Simple Exponential smoothing where only the observation is considered in this decay parameter called alpha for the level. This model is used when the data exhibits no visible trend or seasonality.
- Dobule Exponential smoothing where the trend component is also considered for smoothening. In this case we will have two variables alpha for the level and beta for the trend. . This model is used when the data exhibits only trend but not seasonality. .
- Triple exponential smoothing where all the three components such as level, trend and Seasonality is considered for decaying. Thus leading to alpha for level beta for the trend and gamma for the seasonal component. This model is used when the data exhibit both trend and as well as seasonality.

- Based on the growth of the seasonality in the model. If there is relative growth in the seasonal component the Multiplicative Triple exponential smoothing is used. If the seasonal component is relatively same through out the time series then Triple exponential smoothing with additive method is followed.

For the understanding purpose let us check all these in our model and let us check which model has a better performance based on the RMSE on the test data.

1.4.1.1 Exponential smoothing in Spark Data

Figure 12: Exponential Smoothening comparison plot- Spark Data

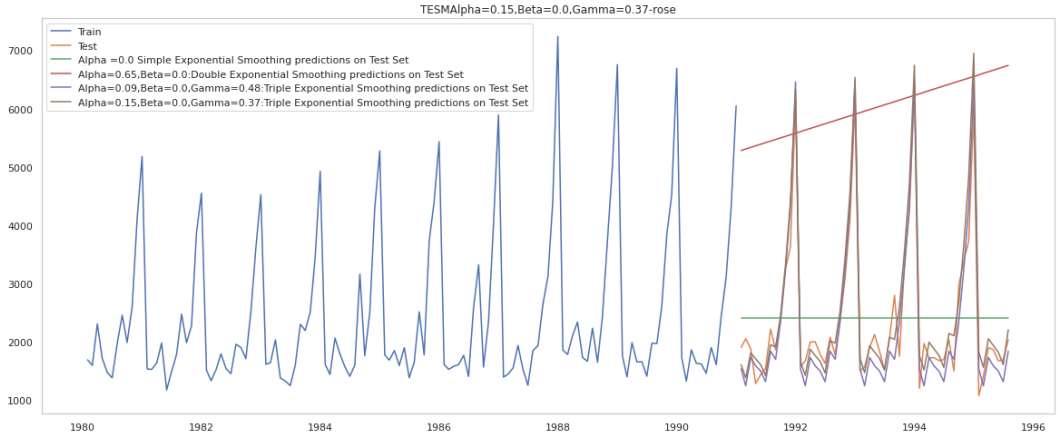


Table 1: Comparison Table- Exponential Smoothing- Spark

	Alpha	Beta	Gamma	Test RMSE
SES- Alpha =0.0-Spark	0.000000	NaN	NaN	1275.081813
DES- Alpha=0.65,Beta=0.0-Spark	0.647814	0.000000e+00	NaN	3850.779835
TES- Alpha=0.09,Beta=0.0,Gamma=0.48- Spark	0.086221	1.054899e-09	0.476416	362.806115
TESMAlpha=0.15,Beta=0.0,Gamma=0.37-Spark	0.154438	7.413317e-28	0.371169	384.197750

We could infer from the Figure 12 and as well as the Table 1, the simple exponential smoothing and double exponential smoothing (without considering the seasonality) has a significantly higher RMSE compared to the triple exponential smoothing. Also it is evident since the data perform best in the additive model.

1.4.1.2 Exponential smoothing in Rose Data

Figure 14: Exponential Smoothening comparison plot- Rose Data

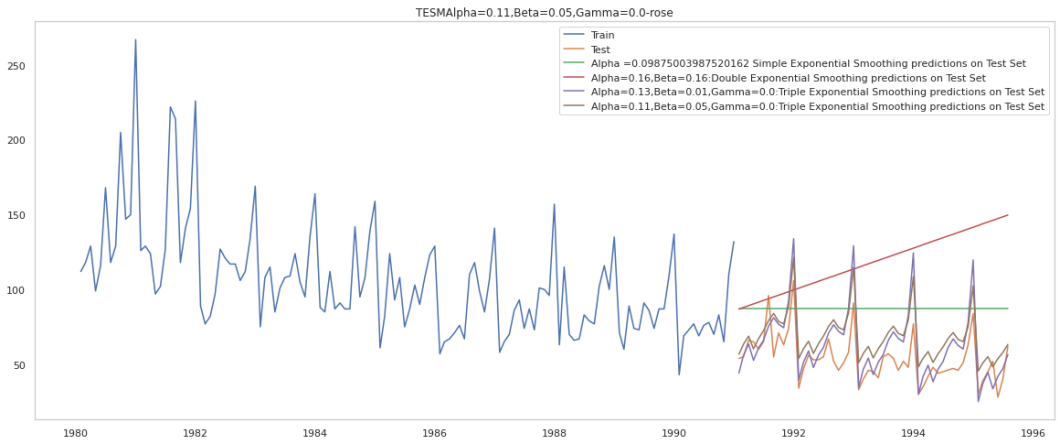


Table 4 Comparison Table- Exponential Smoothing- Rose

	Alpha	Beta	Gamma	Test RMSE
SES- Alpha =0.099-rose	0.098750	NaN	NaN	36.748165
DES- Alpha=0.158,Beta=0.158-rose	0.157895	0.157895	NaN	70.517103
TES- Alpha=0.13,Beta=0.01,Gamma=0.0- rose	0.133470	0.013794	0.0	16.396129
TESMAlpha=0.11,Beta=0.05,Gamma=0.0-rose	0.106096	0.048438	0.0	17.310540

For the Rose data, we could see form Figure 14, and Table 3, there is not much of a difference between Triple Exponential smoothing as Multiplicative and Additive as the Gamma turns out to be zero in both the cases. However, if considered with Simple and double exponential smoothing. The TES model performed well.

1.4.1.3 Other model Comparison.

Although the data is time series we can check few models such as linear regression and Naïve Based model along with simple and moving average models.

1.4.1.3.1 Linear Regression

For a regular linear regression model, we should have an dependent variable and an independent variable. We have considered our sales as independent variable and we have introduced a dependent variable, numbers running along side of the sales in Table 5.

Table 7: Creation of Dependent Variable for Linear Regression

```
First few rows of Training Data
            Sparkling  time
Time_Stamp
1980-01-31       1686     1
1980-02-29       1591     2
1980-03-31       2304     3
1980-04-30       1712     4
1980-05-31       1471     5

Last few rows of Training Data
            Sparkling  time
Time_Stamp
1990-08-31       1605   128
1990-09-30       2424   129
1990-10-31       3116   130
1990-11-30       4286   131
1990-12-31       6047   132
```

```
First few rows of Test Data
            Sparkling  time
Time_Stamp
1991-01-31       1902   133
1991-02-28       2049   134
1991-03-31       1874   135
1991-04-30       1279   136
1991-05-31       1432   137

Last few rows of Test Data
            Sparkling  time
Time_Stamp
1995-03-31       1897   183
1995-04-30       1862   184
1995-05-31       1670   185
1995-06-30       1688   186
1995-07-31       2031   187
```

```
First few rows of Training Data
              Rose  time
Time_Stamp
1980-01-31  112.0     1
1980-02-29  118.0     2
1980-03-31  129.0     3
1980-04-30   99.0     4
1980-05-31  116.0     5

Last few rows of Training Data
              Rose  time
Time_Stamp
1990-08-31   70.0   128
1990-09-30   83.0   129
1990-10-31   65.0   130
1990-11-30  110.0   131
1990-12-31  132.0   132
```

```
First few rows of Test Data
             Rose  time
Time_Stamp
1991-01-31  54.0   133
1991-02-28  55.0   134
1991-03-31  66.0   135
1991-04-30  65.0   136
1991-05-31  60.0   137

Last few rows of Test Data
             Rose  time
Time_Stamp
1995-03-31  45.0   183
1995-04-30  52.0   184
1995-05-31  28.0   185
1995-06-30  40.0   186
1995-07-31  62.0   187
```

Let us see how the linear regression would predict the time series. As they will give an linear relationship on the model this might yield higher RMSE.

1.4.1.3.2 Naïve Bayes Model

One of the models called as Naïve Based model which works based on the previous observation. Here in our case the last observation of our train data becomes the prediction.

1.4.1.3.3 Simple & Moving Average Model

Simple Average and Moving Average are the exclusive methods for the timeseries. Where simple average works by taking the average of the whole train data set leading to only one value for the test data as a prediction. However, in moving average, we can take number of observations before and subtract to predict the current value.

The plots in Figure 17 and RMSE values are in Table 9 of the models are shown b.

For Spark Data:

Figure 17: Comparison plot - Spark Data

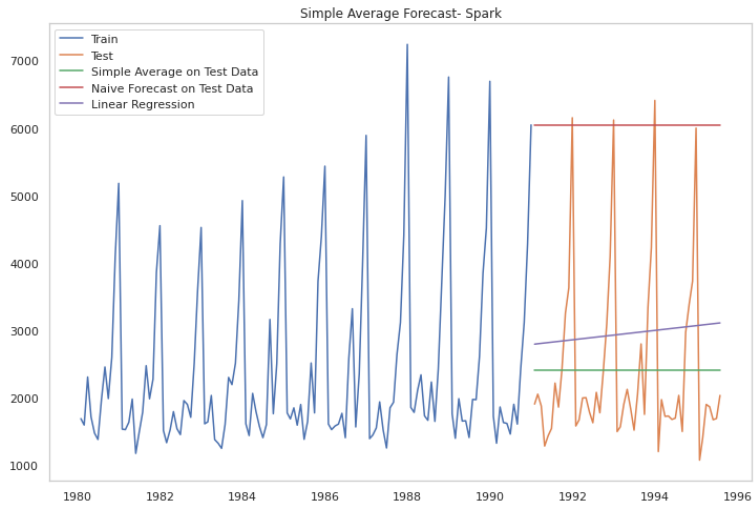
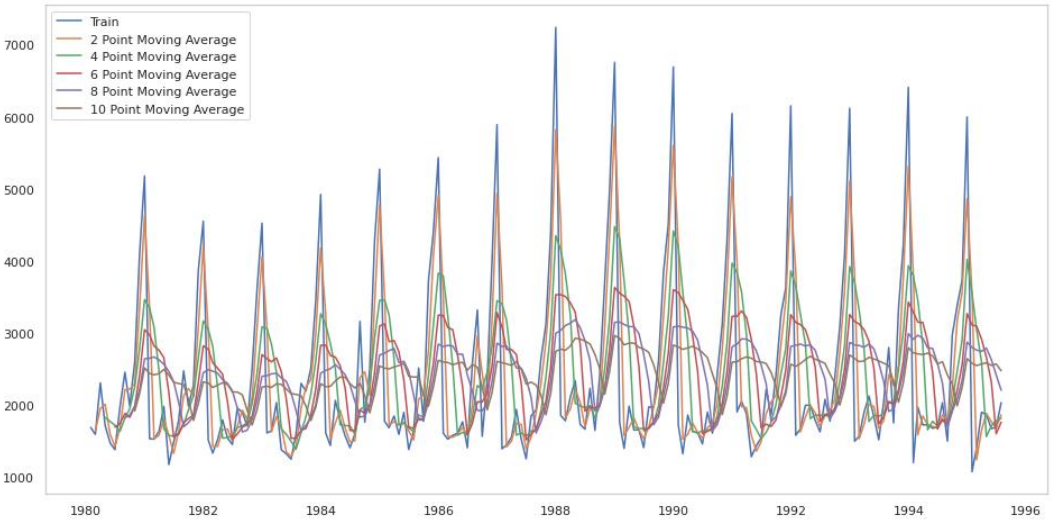


Table 10: Comparison Table- Spark Data

	Alpha	Beta	Gamma	Test RMSE
TES- Alpha=0.09,Beta=0.0,Gamma=0.48- Spark	0.0862207	1.0549e-09	0.476416	362.806115
TESMAlpha=0.15,Beta=0.0,Gamma=0.37-Spark	0.154438	7.41332e-28	0.371169	384.197750
2pointTrailingMovingAverage-spark	-	-	-	813.400684
4pointTrailingMovingAverage-spark	-	-	-	1156.589694
Simple Average Forecast- Spark	-	-	-	1275.081804
SES- Alpha =0.0-Spark	0	NaN	NaN	1275.081813
6pointTrailingMovingAverage-spark	-	-	-	1283.927428
10pointTrailingMovingAverage-spark	-	-	-	1337.748838
8pointTrailingMovingAverage-spark	-	-	-	1342.567772
RegressionOnTime-Spark	-	-	-	1389.135175
DES- Alpha=0.65,Beta=0.0-Spark	0.647814	0	NaN	3850.779835
Naive Forecast-Spark	-	-	-	3864.279352

Figure 20: Moving Average Plot- Spark Data



It is evident from the above graph and the table that TES performs the best in the Spark data.

For Rose Data:

Figure 23: Comparison plot - Rose Data

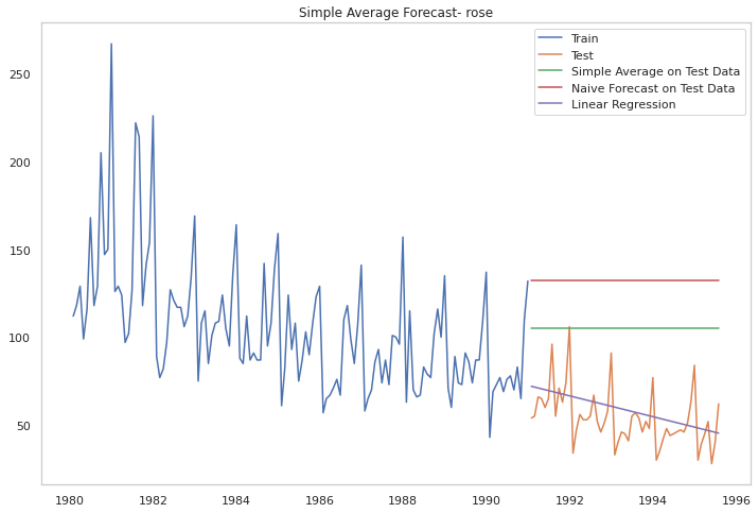
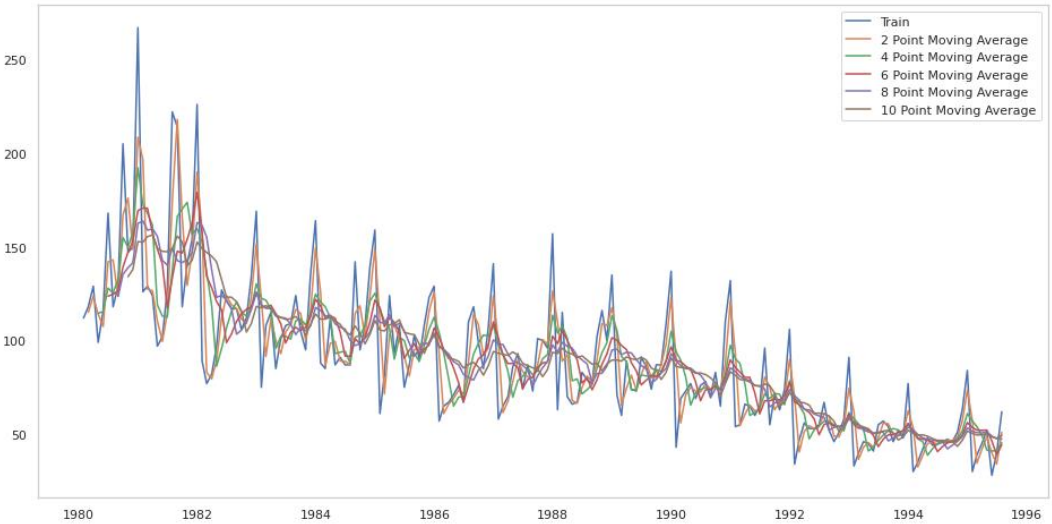


Table 12:Comparison Table- Rose Data

	Alpha	Beta	Gamma	Test RMSE
2pointTrailingMovingAverage-rose	-	-	-	11.529994
4pointTrailingMovingAverage-rose	-	-	-	14.444342
6pointTrailingMovingAverage-rose	-	-	-	14.554944
8pointTrailingMovingAverage-rose	-	-	-	14.806014
10pointTrailingMovingAverage-rose	-	-	-	14.944939
RegressionOnTime-rose	-	-	-	15.255435
TES- Alpha=0.13,Beta=0.01,Gamma=0.0- rose	0.13347	0.0137936	0	16.396129
TESMAlpha=0.11,Beta=0.05,Gamma=0.0-rose	0.106096	0.0484385	0	17.310540
SES- Alpha =0.099-rose	0.09875	NaN	NaN	36.748165
Simple Average Forecast- rose	-	-	-	53.413057
DES- Alpha=0.158,Beta=0.158-rose	0.157895	0.157895	NaN	70.517103
Naive Forecast-rose	-	-	-	79.672238

Figure 25:Moving Average Plot- Rose Data



1.5 Check for the stationarity of the data on which the model is being built on using appropriate statistical tests and mention the hypothesis for the statistical test. If the data is found to be non-stationary, take appropriate steps to make it stationary. Check the new data for stationarity and comment. Note: Stationarity should be checked at alpha = 0.05.

To build the ARIMA Model on to a data, it should be stationary. Meaning the data should have constant mean, constant standard deviation, and no seasonality. However, our data has a seasonality. There are visual ways to see if the data is stationary or not however, ADF (Augmented Dickey Fuller Test).

In this Hypothesis testing

•Null Hypothesis H0: Time Series is non-stationary

•Alternate Hypothesis Ha: Time Series is stationary

If the series is not stationary, then we need to take the differences in the data and the ADF test has to be repeated until we reject the null hypothesis.

For the Spark Data, with the alpha as 0.05,

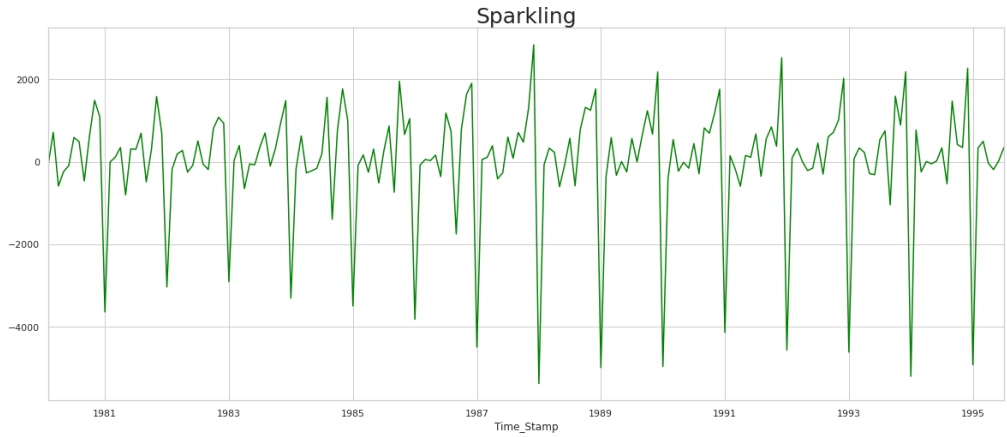
First level of ADF has the pvalue of 0.7055958459932263 and lags used is 12.

When taken a difference of 1 level p value is 0 and lags used is 10.

Hence we reject the null Hypothesis with difference of 1 this also defines the value of d in the ARIMA model.

The Figure 26 shows the data with 1 difference.

Figure 26: Differenced Spark Data



For the Rose Data, with the alpha as 0.05,

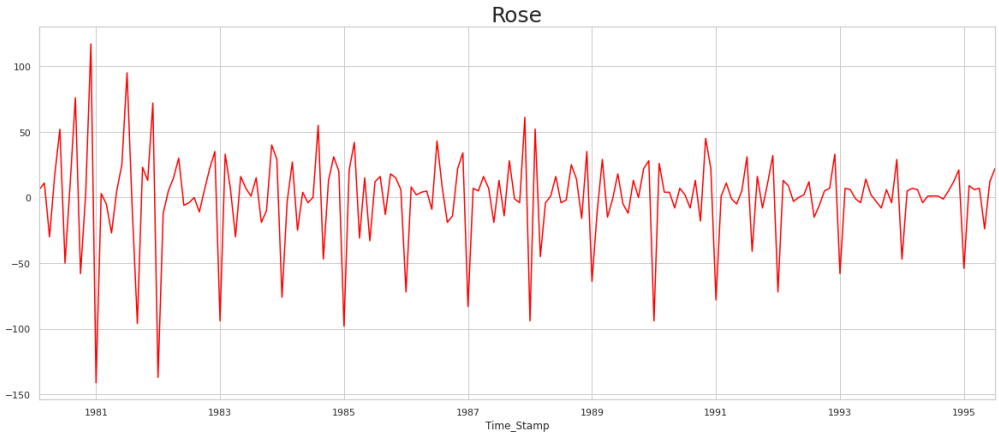
First level of ADF has the pvalue of 0.4675539243113125and lags used is 13.

When taken a difference of 1 level p value is 2.990289942973024e-11 and lags used is 12.

Hence we reject the null Hypothesis with difference of 1 this also defines the value of d in the ARIMA model.

The Figure 27 shows the data with 1 difference.

Figure 27: Differenced Rose Data



1.6 Build an automated version of the ARIMA/SARIMA model in which the parameters are selected using the lowest Akaike Information Criteria (AIC) on the training data and evaluate this model on the test data using RMSE.

After understanding the stationarity, now after bringing the series to stationary or assigning the value of d as 1 we can use brute force method of having definite set of pairs of p, d, q.

- p: represents autoregressive order
- d: represents differencing order
- q: represents moving average order

Iterating the values of p d and q and create the ARIMA Model for each of the pairs of pdq to form the automated model. The AIC (Akaiki Information Criterion) which penalize the number of parameters to the likelihood of the estimation. Leading the lowest AIC model the best. These Automated models will be created and the corresponding AICs is compared and model with the best AIC is chosen.

The code execution file is done in the google colab, where the SARIMAX algorithm is called for both the ARIMA and SARIMA model. Leaving the Seasonal orders, the Algorithm would still work as a ARIMA Model.

With this Assumptions and understanding the ARIMA Model is build with p=(0,1,2,3,4) and q=(0,1,2,3,4) and d= (1)

1.6.1 Automated ARIMA Model

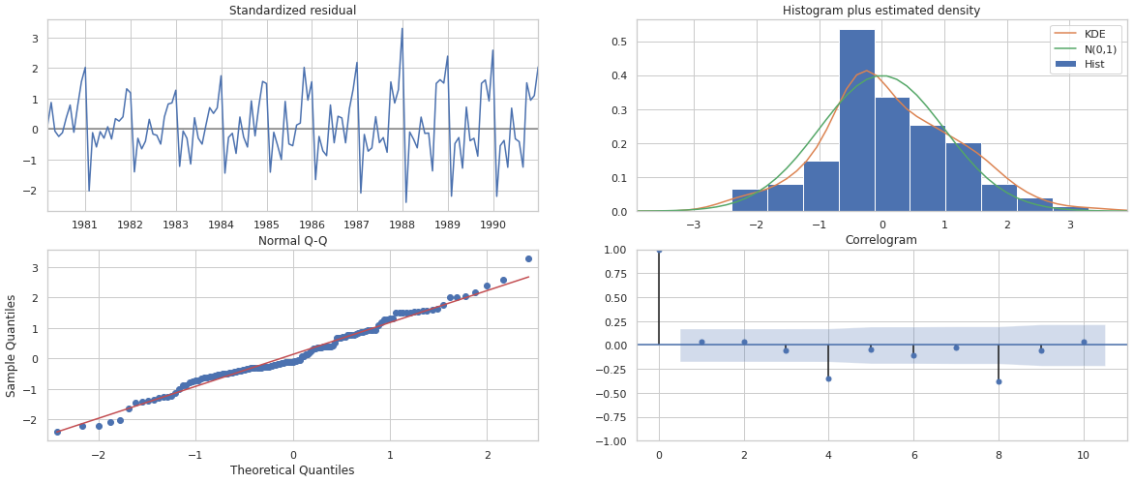
1.6.1.1 Spark Data:

Upon running the Spark train data for the above mentioned iterative numbers the pair of (4,1,4) appears to be having smaller AIC. Their corresponding Results and Diagnostics plot is shown Figure 28

Figure 28: Automated ARIMA-Spark

	param	AIC
0	(4, 1, 4)	2213.036087
1	(2, 1, 2)	2213.509212
2	(3, 1, 4)	2219.893381
3	(3, 1, 3)	2221.437583
4	(4, 1, 3)	2222.904099
5	(2, 1, 4)	2222.921832
6	(1, 1, 4)	2227.736977
7	(3, 1, 2)	2230.827966
8	(2, 1, 3)	2232.972016
9	(4, 1, 2)	2233.061931
10	(2, 1, 1)	2233.777626

```
                               Statespace Model Results
==============================================================================
Dep. Variable:              Sparkling   No. Observations:                  132
Model:               SARIMAX(4, 1, 4)   Log Likelihood               -1097.518
Date:                Sun, 16 Jan 2022   AIC                           2213.036
Time:                        12:49:37   BIC                           2238.913
Sample:                    01-31-1980   HQIC                          2223.551
                         - 12-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.4407      0.106     -4.159      0.000      -0.648      -0.233
ar.L2         -0.4434      0.072     -6.161      0.000      -0.584      -0.302
ar.L3         -0.4430      0.086     -5.128      0.000      -0.612      -0.274
ar.L4          0.5559      0.066      8.414      0.000       0.426       0.685
ma.L1         -0.0067      9.247     -0.001      0.999     -18.130      18.117
ma.L2          0.0122     18.488      0.001      0.999     -36.223      36.247
ma.L3         -0.0182      9.008     -0.002      0.998     -17.673      17.636
ma.L4         -0.9874      0.162     -6.098      0.000      -1.305      -0.670
sigma2      9.084e+05   2.78e-06   3.27e+11      0.000    9.08e+05    9.08e+05
===================================================================================
Ljung-Box (Q):                      338.21   Jarque-Bera (JB):                 0.81
Prob(Q):                              0.00   Prob(JB):                         0.67
Heteroskedasticity (H):               2.82   Skew:                             0.19
Prob(H) (two-sided):                  0.00   Kurtosis:                         3.09
===================================================================================
```



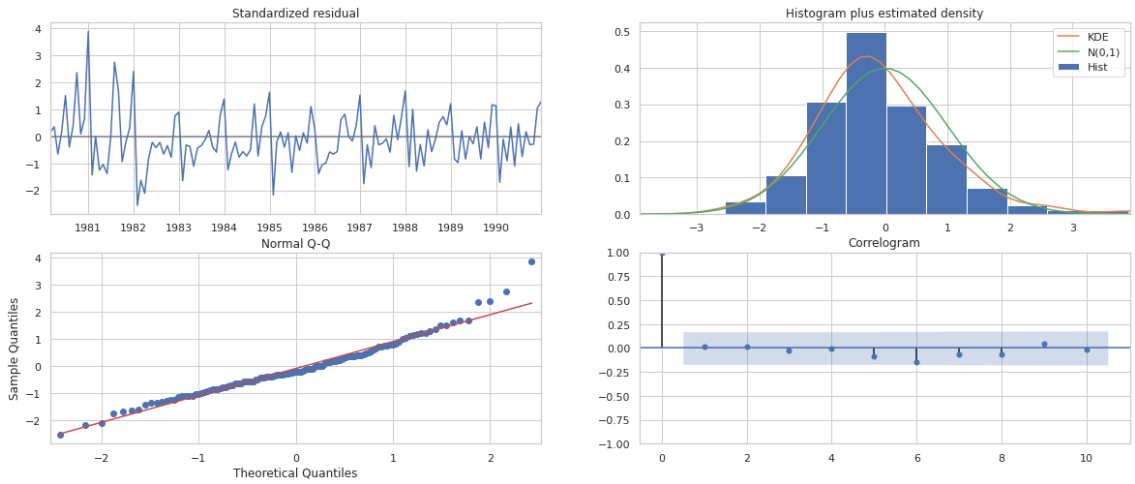
1.6.1.2 Rose Data:

Upon running the Rose train data for the above mentioned iterative numbers the pair of (2,1,3) appears to be having smaller AIC. Their corresponding Results and Diagnostics plot is shown Figure 29

Figure 29 :Automated ARIMA-Rose

	param	AIC
0	(2, 1, 3)	1274.695818
1	(4, 1, 3)	1278.451416
2	(3, 1, 3)	1278.664523
3	(2, 1, 4)	1278.772088
4	(1, 1, 4)	1279.605262
5	(0, 1, 2)	1279.671529
6	(1, 1, 2)	1279.870723
7	(0, 1, 3)	1280.545376
8	(1, 1, 1)	1280.574230
9	(2, 1, 1)	1281.507862
10	(0, 1, 4)	1281.676698

```
                           Statespace Model Results
==============================================================================
Dep. Variable:                   Rose   No. Observations:                  132
Model:               SARIMAX(2, 1, 3)   Log Likelihood                -631.348
Date:                Sun, 16 Jan 2022   AIC                           1274.696
Time:                        12:49:45   BIC                           1291.947
Sample:                    01-31-1980   HQIC                          1281.706
                         - 12-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -1.6773      0.084    -19.988      0.000      -1.842      -1.513
ar.L2         -0.7281      0.084     -8.673      0.000      -0.893      -0.564
ma.L1          1.0458      0.673      1.554      0.120      -0.273       2.364
ma.L2         -0.7700      0.137     -5.611      0.000      -1.039      -0.501
ma.L3         -0.9037      0.610     -1.482      0.138      -2.099       0.292
sigma2       859.2102    567.079      1.515      0.130    -252.245    1970.665
===================================================================================
Ljung-Box (Q):                      101.06   Jarque-Bera (JB):                24.19
Prob(Q):                              0.00   Prob(JB):                         0.00
Heteroskedasticity (H):               0.40   Skew:                             0.70
Prob(H) (two-sided):                  0.00   Kurtosis:                         4.56
===================================================================================
```



1.6.2 Automated SARIMA Model

For Automated SARIMA Model we have additional orders Accounting for the seasonality. They are

- P: seasonal autoregressive order
- D: seasonal differencing order
- Q: seasonal moving average order
- S: Number of time steps per cycle

Here S represents the seasonal Data span which is 12 months. Also, the PDQ are the corresponding Seasonal entity for AR I and MA. Since the iteration has to go through (p,d,q) and (P,D,Q, S) the data would take substantially higher running time hence the p,q,P,Q are considered only between (0,1,2).

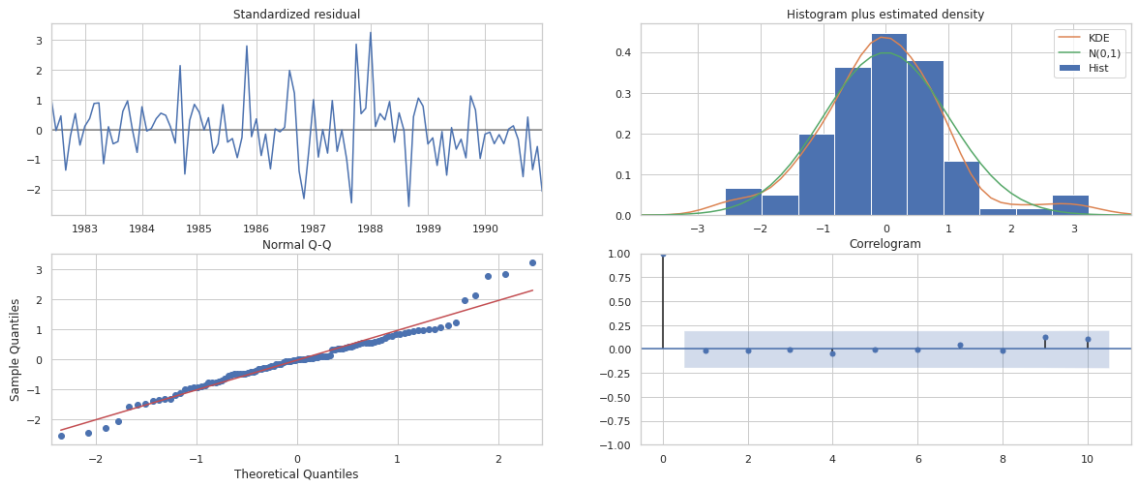
1.6.2.1 Spark Data

Upon running the Spark train data for the above-mentioned iterative numbers the pair of (1,1,2) (1,0,2,12) appears to be having smaller AIC. Their corresponding Results and Diagnostics plot is shown Figure 30

Figure 30: Automated SARIMA-Spark

	param	seasonal	AIC
0	(1, 1, 2)	(1, 0, 2, 12)	1555.584247
1	(1, 1, 2)	(2, 0, 2, 12)	1555.934564
2	(0, 1, 2)	(2, 0, 2, 12)	1557.121563
3	(0, 1, 2)	(1, 0, 2, 12)	1557.160507
4	(2, 1, 2)	(1, 0, 2, 12)	1557.340409
5	(2, 1, 2)	(2, 0, 2, 12)	1557.840130
6	(0, 1, 1)	(1, 0, 2, 12)	1570.150914
7	(0, 1, 1)	(2, 0, 2, 12)	1570.401882
8	(1, 1, 1)	(1, 0, 2, 12)	1571.324886
9	(1, 1, 1)	(2, 0, 2, 12)	1571.806997
10	(2, 1, 1)	(1, 0, 2, 12)	1571.988828

```
                                     Statespace Model Results
==========================================================================================
Dep. Variable:                              Sparkling   No. Observations:                  132
Model:                 SARIMAX(1, 1, 2)x(1, 0, 2, 12)   Log Likelihood                -770.792
Date:                                Sun, 16 Jan 2022   AIC                           1555.584
Time:                                        12:51:04   BIC                           1574.095
Sample:                                    01-31-1980   HQIC                          1563.083
                                         - 12-31-1990
Covariance Type:                                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.6282      0.255     -2.464      0.014      -1.128      -0.128
ma.L1         -0.1040      0.225     -0.463      0.643      -0.545       0.336
ma.L2         -0.7277      0.154     -4.736      0.000      -1.029      -0.427
ar.S.L12       1.0439      0.014     72.837      0.000       1.016       1.072
ma.S.L12      -0.5550      0.098     -5.663      0.000      -0.747      -0.363
ma.S.L24      -0.1354      0.120     -1.133      0.257      -0.370       0.099
sigma2      1.506e+05   2.03e+04      7.401      0.000    1.11e+05     1.9e+05
===================================================================================
Ljung-Box (Q):                       23.02   Jarque-Bera (JB):                11.72
Prob(Q):                              0.99   Prob(JB):                         0.00
Heteroskedasticity (H):               1.47   Skew:                             0.36
Prob(H) (two-sided):                  0.26   Kurtosis:                         4.48
===================================================================================
```

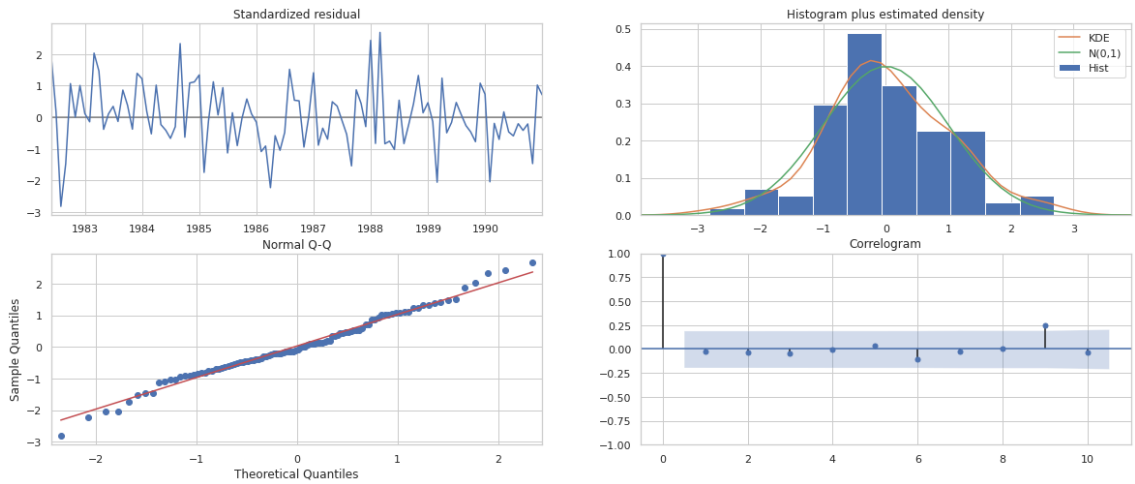


1.6.2.2 Rose Data

Upon running the Rose train data for the above-mentioned iterative numbers the pair of (0,1,2) (2,0,2,12) appears to be having smaller AIC. Their corresponding Results and Diagnostics plot is shown below

	param	seasonal	AIC
0	(0, 1, 2)	(2, 0, 2, 12)	887.937509
1	(1, 1, 2)	(2, 0, 2, 12)	889.900383
2	(2, 1, 2)	(2, 0, 2, 12)	890.668798
3	(2, 1, 1)	(2, 0, 0, 12)	896.518161
4	(2, 1, 2)	(2, 0, 0, 12)	897.346444
5	(2, 1, 1)	(2, 0, 1, 12)	897.639957
6	(2, 1, 2)	(2, 0, 1, 12)	898.378189
7	(2, 1, 1)	(2, 0, 2, 12)	899.483587
8	(1, 1, 1)	(2, 0, 2, 12)	900.672580
9	(0, 1, 1)	(2, 0, 2, 12)	901.198826
10	(0, 1, 2)	(1, 0, 2, 12)	904.831091

```
                                     Statespace Model Results
==========================================================================================
Dep. Variable:                                 Rose   No. Observations:                  132
Model:                 SARIMAX(1, 1, 2)x(1, 0, 2, 12)   Log Likelihood                -446.366
Date:                                Sun, 16 Jan 2022   AIC                            906.732
Time:                                        12:51:59   BIC                            925.243
Sample:                                    01-31-1980   HQIC                           914.231
                                         - 12-31-1990
Covariance Type:                                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.1141      0.369     -0.310      0.757      -0.837       0.608
ma.L1         -0.6699    152.458     -0.004      0.996    -299.482     298.142
ma.L2         -0.3301     50.385     -0.007      0.995     -99.083      98.422
ar.S.L12       0.6255      0.059     10.548      0.000       0.509       0.742
ma.S.L12      -0.1614      0.126     -1.285      0.199      -0.407       0.085
ma.S.L24       0.1133      0.134      0.844      0.399      -0.150       0.376
sigma2       299.9909   4.58e+04      0.007      0.995   -8.94e+04       9e+04
===================================================================================
Ljung-Box (Q):                       24.41   Jarque-Bera (JB):                 0.59
Prob(Q):                              0.98   Prob(JB):                         0.74
Heteroskedasticity (H):               0.71   Skew:                             0.08
Prob(H) (two-sided):                  0.31   Kurtosis:                         3.33
===================================================================================
```



1.7 Build ARIMA/SARIMA models based on the cut-off points of ACF and PACF on the training data and evaluate this model on the test data using RMSE.

Earlier we have followed brute force method to define the pdq values, as we have a leverage of computational power. However, in the past, this is done by looking at Auto Correlation Function (ACF) and Partial Auto Correlation Function (PACF) plots.

ACF- Shows the value of q and PACF shows the value of p.

Plot details:

The plot has a band of 0.05 alpha value meaning the confidence level band of 95%.

Lags from the original data is plotted in the x axis against the percentage of correlation.

The ACF(0) is always 1.0 as the original data is always 100% correlated to itself.

Similarly PACF(1)= ACF(1).

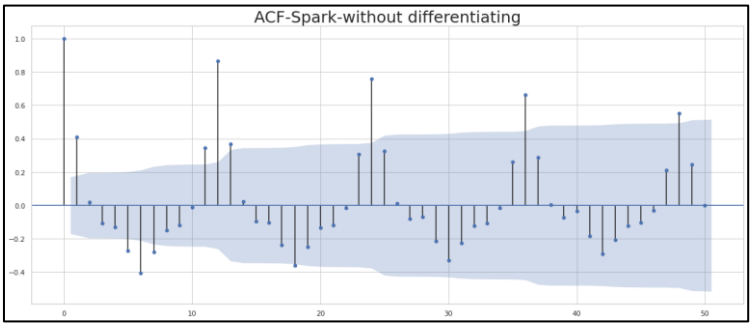
Finding the significant number:

The significant lags are the ones which are situated outside the confidence band. And we count the number of significant bands from 0.

Here, since the data was not stationary**, the PACF and ACF Plots are done for the first difference of the data.**

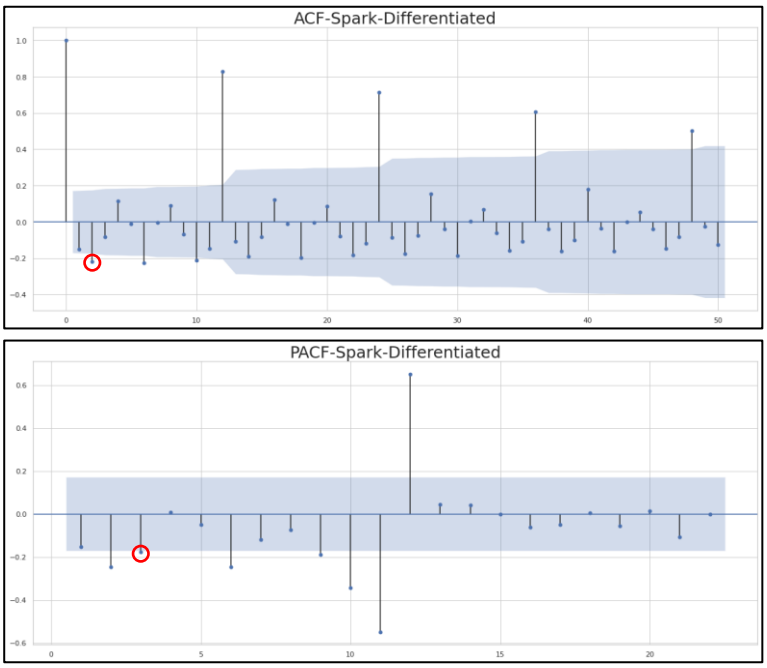
If the plots are created for the non-stationary data ie the Seasonal Data, the ACF plots will also reflect the seasonal components as shown Figure 31.

Figure 31: ACF plot showing Seasonality



1.7.1 Spark Data

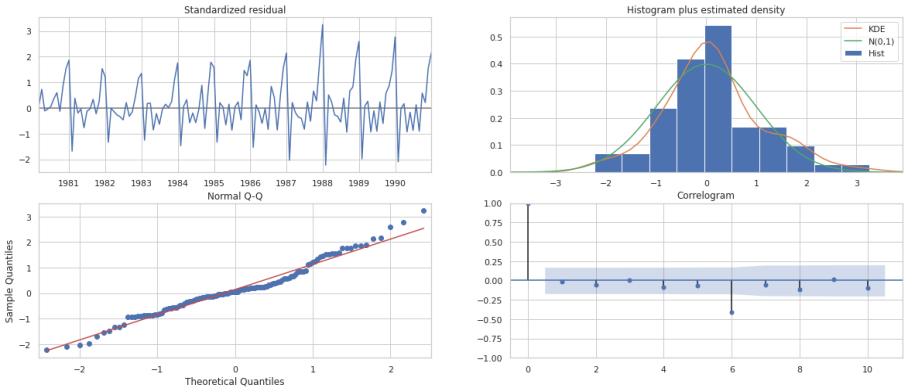
Figure 32: ACF and PCF Plots of Spark Data



After looking into the Figure 32 we could see p, d,and q can be chosen as (3,1,2) and the corresponding Statistics and Diagnostics plots are as shown in Figure 33.

Figure 33: Manual ARIMA-Spark

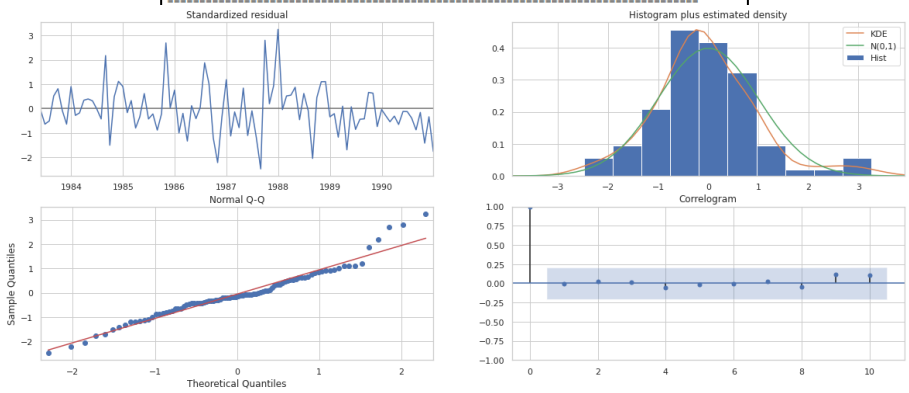
```
                           Statespace Model Results
==============================================================================
Dep. Variable:              Sparkling   No. Observations:                  132
Model:               SARIMAX(3, 1, 2)   Log Likelihood               -1109.414
Date:                Sun, 16 Jan 2022   AIC                           2230.828
Time:                        16:13:09   BIC                           2248.079
Sample:                    01-31-1980   HQIC                          2237.838
                         - 12-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.4338      0.043    -10.010      0.000      -0.519      -0.349
ar.L2          0.3312      0.106      3.132      0.002       0.124       0.538
ar.L3         -0.2350      0.062     -3.762      0.000      -0.357      -0.113
ma.L1          0.0136      0.131      0.104      0.917      -0.243       0.271
ma.L2         -0.9862      0.138     -7.148      0.000      -1.257      -0.716
sigma2      1.283e+06   1.97e-07   6.52e+12      0.000    1.28e+06    1.28e+06
===================================================================================
Ljung-Box (Q):                      331.01   Jarque-Bera (JB):                 5.05
Prob(Q):                              0.00   Prob(JB):                         0.08
Heteroskedasticity (H):               2.73   Skew:                             0.38
Prob(H) (two-sided):                  0.00   Kurtosis:                         3.58
===================================================================================
```



Similarly for the SARIMA Model (p,d,q) and (P,D,Q,S) are chosen as (2,1,2) and (2,1,2,12).

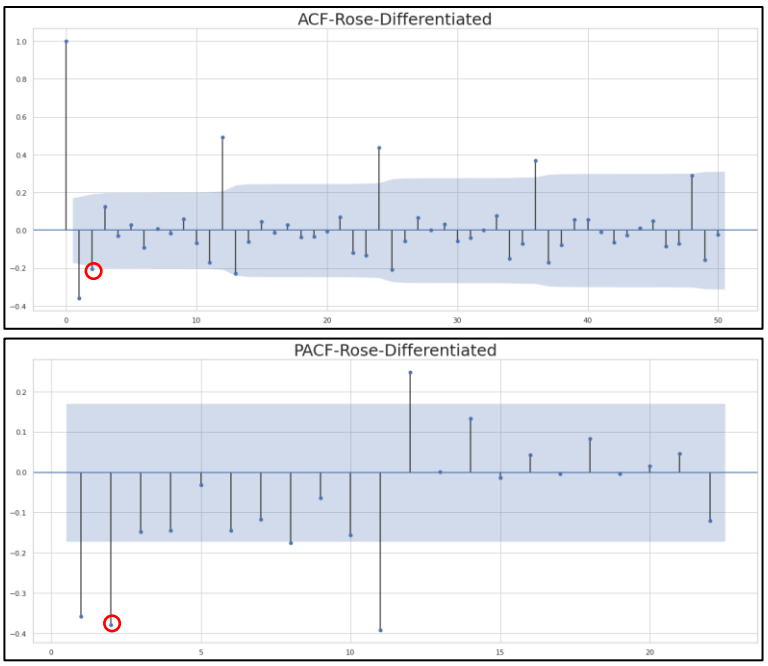
Figure 34: Manual SARIMA-Spark

```
                                     Statespace Model Results
==========================================================================================
Dep. Variable:                                Sparkling   No. Observations:                  132
Model:                   SARIMAX(2, 1, 2)x(2, 1, 2, 12)   Log Likelihood                -684.814
Date:                                  Sun, 16 Jan 2022   AIC                           1387.628
Time:                                          16:13:18   BIC                           1410.324
Sample:                                      01-31-1980   HQIC                          1396.788
                                           - 12-31-1990
Covariance Type:                                    opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.5623      0.332     -1.695      0.090      -1.212       0.088
ar.L2          0.0325      0.167      0.195      0.845      -0.294       0.359
ma.L1         -0.1506      0.293     -0.513      0.608      -0.725       0.424
ma.L2         -0.7626      0.259     -2.942      0.003      -1.271      -0.254
ar.S.L12      -0.2455      0.977     -0.251      0.802      -2.161       1.670
ar.S.L24      -0.1093      0.215     -0.508      0.612      -0.532       0.313
ma.S.L12      -0.1621      0.976     -0.166      0.868      -2.076       1.752
ma.S.L24      -0.0210      0.440     -0.048      0.962      -0.884       0.842
sigma2        1.7e+05   2.53e+04      6.731      0.000    1.21e+05     2.2e+05
===================================================================================
Ljung-Box (Q):                       22.34   Jarque-Bera (JB):                13.51
Prob(Q):                              0.99   Prob(JB):                         0.00
Heteroskedasticity (H):               0.95   Skew:                             0.59
Prob(H) (two-sided):                  0.88   Kurtosis:                         4.46
===================================================================================
```



1.7.2 Rose Data

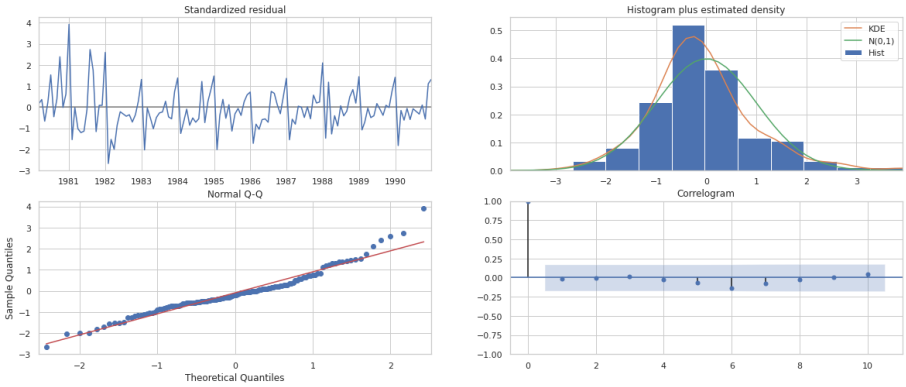
Figure 35: ACF and PACF plot of Rose Data



After looking into the Figure 35 we could see p, d,and q can be chosen as (2,1,2) and the corresponding Statistics and Diagnostics plots are as shown Figure 36.

Figure 36:Manual ARIMA-Rose

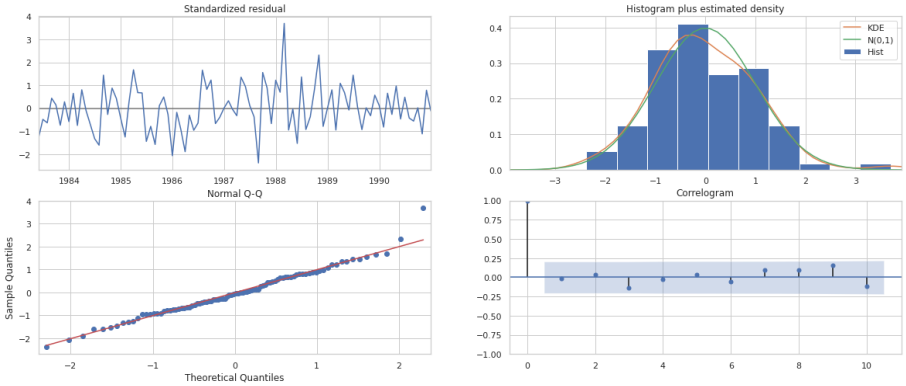
```
                               Statespace Model Results
==============================================================================
Dep. Variable:                   Rose   No. Observations:                  132
Model:               SARIMAX(2, 1, 2)   Log Likelihood                -635.935
Date:                Sun, 16 Jan 2022   AIC                           1281.871
Time:                        16:13:12   BIC                           1296.247
Sample:                    01-31-1980   HQIC                          1287.712
                         - 12-31-1990
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.4540      0.469     -0.969      0.333      -1.372       0.464
ar.L2          0.0001      0.170      0.001      0.999      -0.334       0.334
ma.L1         -0.2541      0.459     -0.554      0.580      -1.154       0.646
ma.L2         -0.5984      0.430     -1.390      0.164      -1.442       0.245
sigma2       952.1601     91.424     10.415      0.000     772.973    1131.347
===================================================================================
Ljung-Box (Q):                      112.12   Jarque-Bera (JB):                34.16
Prob(Q):                              0.00   Prob(JB):                         0.00
Heteroskedasticity (H):               0.37   Skew:                             0.79
Prob(H) (two-sided):                  0.00   Kurtosis:                         4.94
===================================================================================
```



Similarly for the SARIMA Model (p,d,q) and (P,D,Q,S) are chosen as (2,1,2) and (2,1,2,12).

Figure 37: Manual SARIMA-Rose

```
                                     Statespace Model Results
==========================================================================================
Dep. Variable:                                Rose   No. Observations:                  132
Model:                   SARIMAX(2, 1, 2)x(2, 1, 2, 12)   Log Likelihood                -379.498
Date:                                  Sun, 16 Jan 2022   AIC                            776.996
Time:                                          16:13:27   BIC                            799.692
Sample:                                      01-31-1980   HQIC                           786.156
                                           - 12-31-1990
Covariance Type:                                    opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.8551      0.146     -5.838      0.000      -1.142      -0.568
ar.L2         -0.0022      0.125     -0.018      0.986      -0.247       0.242
ma.L1          0.0120      0.184      0.066      0.948      -0.348       0.372
ma.L2         -0.9434      0.150     -6.292      0.000      -1.237      -0.650
ar.S.L12       0.0347      0.185      0.188      0.851      -0.328       0.397
ar.S.L24      -0.0459      0.029     -1.599      0.110      -0.102       0.010
ma.S.L12      -0.7224      0.333     -2.172      0.030      -1.374      -0.071
ma.S.L24      -0.0772      0.212     -0.364      0.716      -0.493       0.339
sigma2       192.1778     39.487      4.867      0.000     114.785     269.570
===================================================================================
Ljung-Box (Q):                       29.82   Jarque-Bera (JB):                 7.06
Prob(Q):                              0.88   Prob(JB):                         0.03
Heteroskedasticity (H):               0.87   Skew:                             0.45
Prob(H) (two-sided):                  0.71   Kurtosis:                         4.01
===================================================================================
```



1.8 Build a table (create a data frame) with all the models built along with their corresponding parameters and the respective RMSE values on the test data.

Table 14: Model comparison Table

	Data	Type	AIC	RMSE	MAPE
SARIMA: (2, 1, 2)(2, 1, 2, 12)	Rose	Manual	776.996101	16.508883	25.350957
SARIMA: (0, 1, 2)(2, 0, 2, 12)	Rose	Auto	906.731850	31.428786	64.212882
ARIMA(2, 1, 3)	Rose	Auto	1274.695818	36.787417	75.709320
ARIMA (2,1,2)	Rose	Manual	1281.870722	36.823178	75.879704
SARIMA: (2, 1, 2)(2, 1, 2, 12)	Sparkling	Manual	1387.627785	345.938884	13.104585
SARIMA: (1, 1, 2)(1, 0, 2, 12)	Sparkling	Auto	1555.584247	528.602513	20.955311
ARIMA(4, 1, 4)	Sparkling	Auto	2213.036087	1213.248777	40.275796
ARIMA (3,1,2)	Sparkling	Manual	2230.827966	1278.074959	42.499545

1.9 Based on the model-building exercise, build the most optimum model(s) on the complete data and predict 12 months into the future with appropriate confidence intervals/bands.

From Table 14 we can say SARIMA Manual method is best for both the Spark and Rose Data.

The model can be built with order as SARIMAX(2, 1, 2)x(2, 1, 2, 12) for both the data.

The predicted model is shown in blue color in Figure 39,Figure 41 for 12 months of span for both the data.

Figure 38: Full SARIMA-Sparkling

```
                                     Statespace Model Results
==========================================================================================
Dep. Variable:                                Sparkling   No. Observations:                  187
Model:                   SARIMAX(2, 1, 2)x(2, 1, 2, 12)   Log Likelihood               -1085.734
Date:                                  Sun, 16 Jan 2022   AIC                           2189.469
Time:                                          13:32:14   BIC                           2216.383
Sample:                                      01-31-1980   HQIC                          2200.404
                                           - 07-31-1995
Covariance Type:                                    opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.6607      0.280     -2.363      0.018      -1.209      -0.113
ar.L2         -0.0094      0.114     -0.083      0.934      -0.232       0.213
ma.L1         -0.1728      0.265     -0.652      0.514      -0.692       0.346
ma.L2         -0.7322      0.247     -2.968      0.003      -1.216      -0.249
ar.S.L12       0.0489      0.378      0.130      0.897      -0.692       0.790
ar.S.L24       0.2819      0.137      2.053      0.040       0.013       0.551
ma.S.L12      -0.6954      0.535     -1.300      0.194      -1.744       0.353
ma.S.L24      -0.2193      0.325     -0.674      0.500      -0.857       0.418
sigma2       1.38e+05   3.26e+04      4.237      0.000    7.42e+04    2.02e+05
===================================================================================
Ljung-Box (Q):                       17.15   Jarque-Bera (JB):                26.96
Prob(Q):                              1.00   Prob(JB):                         0.00
Heteroskedasticity (H):               0.88   Skew:                             0.57
Prob(H) (two-sided):                  0.65   Kurtosis:                         4.76
===================================================================================
```

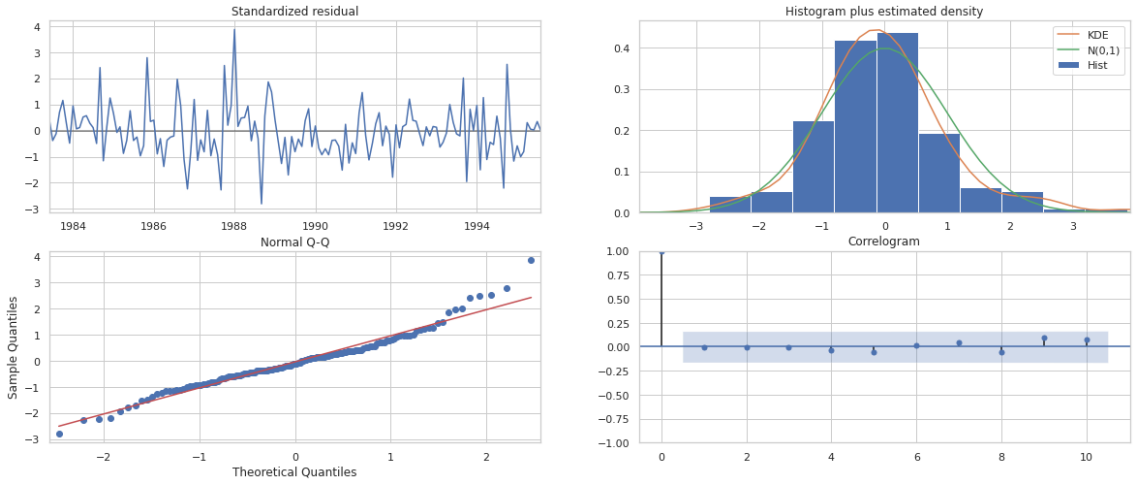
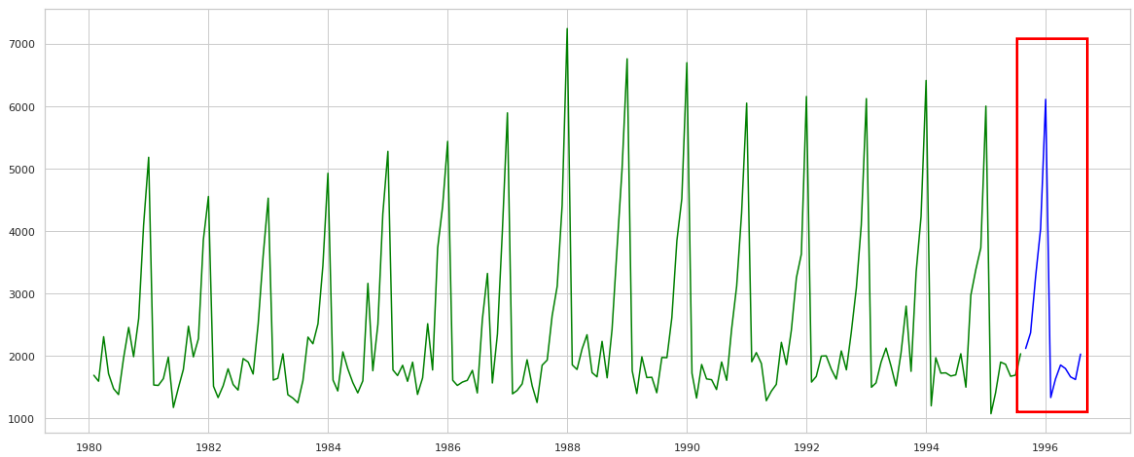


Figure 39: Full Data Prediction- Future-Spark



For Rose Data:

Figure 40: Full SARIMA-Rose

```
                                     Statespace Model Results
==========================================================================================
Dep. Variable:                                   Rose   No. Observations:                  187
Model:             SARIMAX(2, 1, 2)x(2, 1, 2, 12)   Log Likelihood                -587.524
Date:                            Sun, 16 Jan 2022   AIC                           1193.047
Time:                                    13:32:38   BIC                           1219.961
Sample:                                01-31-1980   HQIC                          1203.983
                                     - 07-31-1995
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.8546      0.103     -8.285      0.000      -1.057      -0.652
ar.L2          0.0465      0.093      0.503      0.615      -0.135       0.228
ma.L1          0.0874      0.687      0.127      0.899      -1.259       1.434
ma.L2         -0.9089      0.625     -1.455      0.146      -2.133       0.315
ar.S.L12       0.1082      0.163      0.663      0.507      -0.212       0.428
ar.S.L24      -0.0349      0.016     -2.169      0.030      -0.066      -0.003
ma.S.L12      -0.7348      0.221     -3.325      0.001      -1.168      -0.302
ma.S.L24      -0.0242      0.153     -0.158      0.874      -0.324       0.276
sigma2       160.2412    104.519      1.533      0.125     -44.612     365.094
===================================================================================
Ljung-Box (Q):                       36.11   Jarque-Bera (JB):                10.21
Prob(Q):                              0.65   Prob(JB):                         0.01
Heteroskedasticity (H):               0.54   Skew:                             0.36
Prob(H) (two-sided):                  0.03   Kurtosis:                         4.08
===================================================================================
```

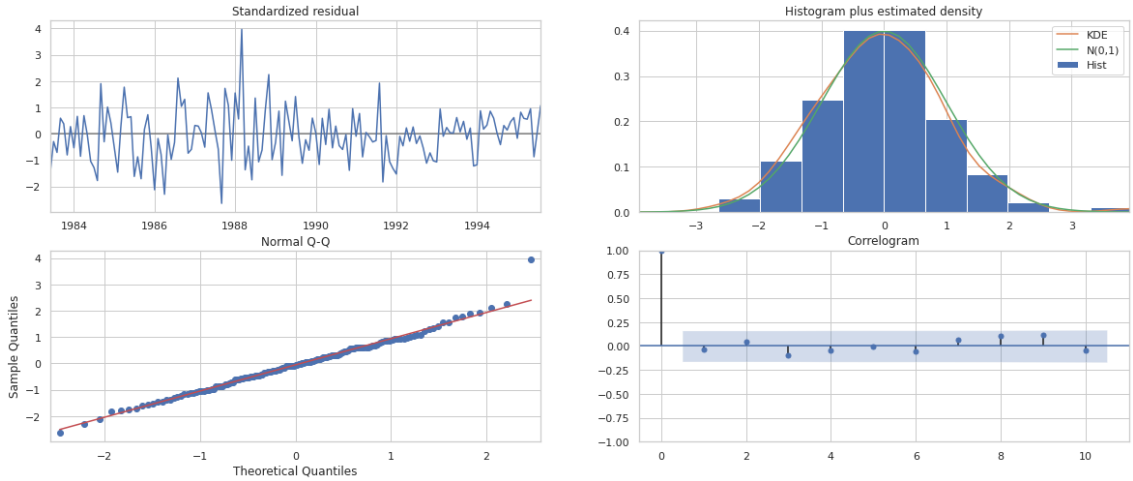
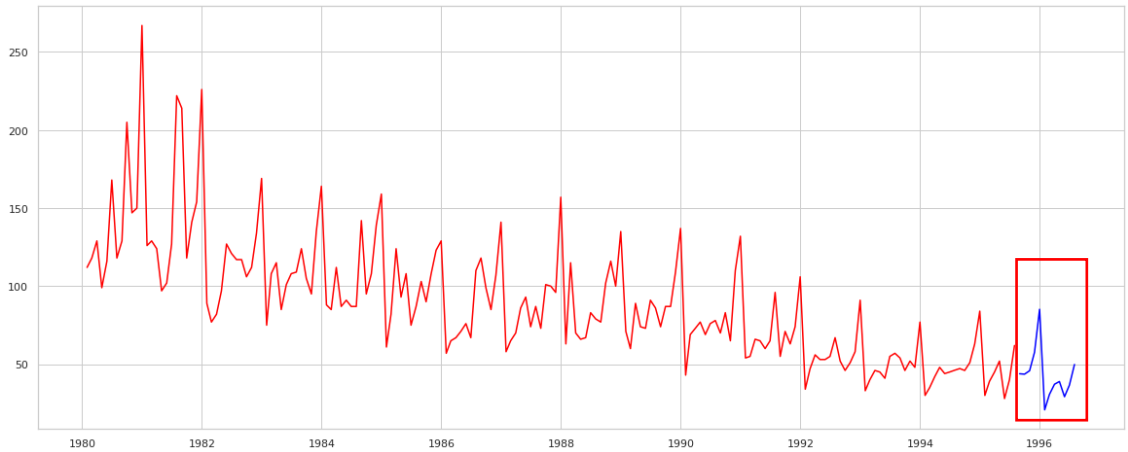


Figure 41:Full_Data Prediction- Future-Rose



1.10 Comment on the model thus built and report your findings and suggest the measures that the company should be taking for future sales.

- The model is then decomposed find weather we have the seasonal component and trend or not. Out of two model Rose exhibit downward trend and Spark exhibit an Hybrid trend.
- Both the data has a seasonal factor. And it is evident that December month's sales are the highest on both the data with respect to their sales scales.
- Both the models have relatively similar seasonal trend and hence we have chosen additive methods for the further calculation.
- Smoothening is done for both the model. Since the series has both trend and Seasonality Triple exponential smoothening was the best option to choose. The results also lead to the same.
- Other than this Naïve model and Linear regressions were also made and was compared with the test data.
- Simple and Moving averages were also performed to predict the data. The two-point average leads to the lowest RMSE value.
- Since there is significance of seasonality, the series should be made stationary before building the model
- ARIMA and SARIMA models were built by brute force method of choosing pdq values (iterative process).
- Later by finding the significant lag on the PCF and ACF Plot ARIMA and SARIMA models were also built.
- Since the model has a significant Seasonal component. The SARIMA models shows a greater performance compared to ARIMA model.

1.10.1 Business Recommendation:

1.10.1.1 Rose:

Since the trend is growing down we should see factors that are leading to the downtrend. To increase the sales may be we can give discounts on the Rose sales at the early year which can boost the sales. April month has the lowest sales, this may be due to change in financial year, if there is a change in price level this may lead to decrease in sales so the selling price change should be avoided at this month.

1.10.1.2 Sparkling:

There are almost no sales in the mid of the year (June month). This may be because the colleges and schools are reopened after vacation also due to the high summer impact. Hence promotions to be given right after summer in such a way we can boost the sales by June. The spark series has high residual compared to Rose. Meaning that there are several reasons behind the fluctuations that could not be capture.